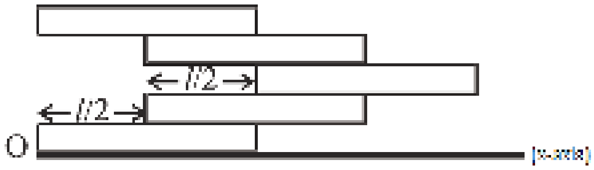
PHYSICS

1) Five identical homogeneous bricks, each of length ℓ, are arranged as shown in figure.X-coordinate of the centre of mass relative to the origin O is :



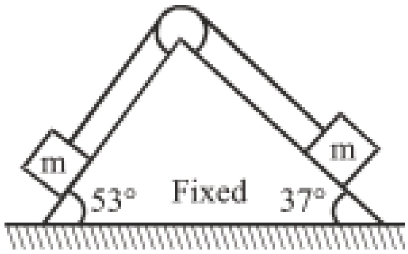
(1) $9\ell/2$

(2) $9\ell/5$

(3) $9\ell/10$

(4) $9\ell/4$

2) Find the magnitude of acceleration of centre of mass of two blocks as shown in figure. Neglect friction everywhere. (g = 10 ms^{-2})



(1) 1 ms^{-2}

(2) $\frac{1}{\sqrt{2}} ms^{-2}$

(3) $\sqrt{2}\, ms^{-1}$

(4) Zero

3) A fast moving neutron collides elastically with stationary α-particle then fraction of retained kinetic energy of neutron is :-

(1) $\frac{16}{25}$

(2) $\frac{9}{25}$

(3) $\frac{13}{25}$

(4) $\frac{12}{15}$

4) Match the column:

Column-I	Column-II

(P)	Perfect elastic collision	(1)	Only Momentum is conserved.
(Q)	Inelastic collision	(2)	Momentum conserved and Kinetic energy is equal before and after collision.
(R)	Perfect inelastic collision	(3)	Momentum conserved KE lost, e = 0
(S)	Explosion of bomb at rest.	(4)	Momentum conserved, KE lost 0 < e < 1
		(5)	Momentum conserved, KE increases

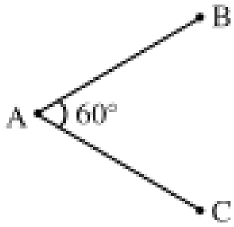
(1) P→2; Q→4; R→3; S→5

(2) P→2; Q→1; R→3; S→5

(3) P→1; Q→4; R→3; S→2

(4) P→2; Q→4; R→1; S→5

5) A uniform wire of length ℓ is bent into v shape. The distance of its centre of mass from vertex A.



(1) $\frac{\ell}{2}$

(2) $\frac{\ell\sqrt{3}}{4}$

(3) $\frac{\ell\sqrt{3}}{8}$

(4) None

6) The speed of a particle moving in a circle slows down at a rate of 3 m/sec^2. At some instant the magnitude of the total acceleration is 5 m/sec^2 and the particle's speed is 12 m/sec. The radius of circle will be :-

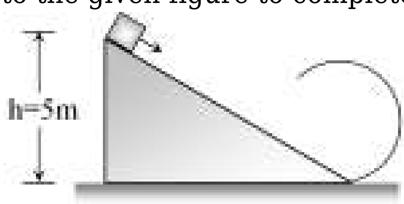
(1) 12 m

(2) 24 m

(3) 36 m

(4) 48 m

7) According to the given figure to complete the circular loop what should be the radius if initial height is 5 m



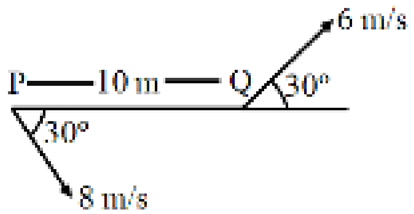
(1) 4 m

(2) 3 m

(3) 2.5 m

(4) 2 m

8) Two moving particles 'P' and 'Q' are 10 m apart at an certain instant. The velocity of P is 8 m/s making an angle 30° with line joining P and Q and velocity of Q is 6 m/s making an angle 30° with PQ as shown in figure. Then the magnitude of angular velocity of P with respect to Q is -



(1) 0 rad/s

(2) 0.1 rad/s

(3) 0.4 rad/s

(4) 0.7 rad/s

9) In a motorcycle stunt called the well of death, the track is vertical cylindrical surface of 36m radius. If minimum speed of the motorcycle to prevent him from sliding down is 30 m/s then what is the coefficient of friction : ($g = 10\ m/s^2$)

(1) 0.2

(2) 0.3

(3) 0.4

(4) $\frac{1}{\sqrt{2}}$

10) Which of the given values of angular speed is possible for which coin does not slip on rotating

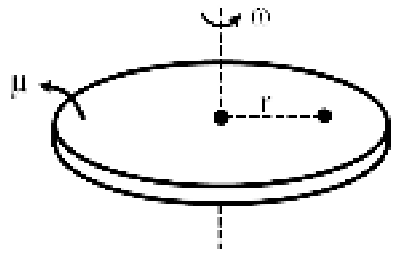


table (given μ = 0.5, r = 10 cm)

(1) 7 rad/s

(2) 3 rad/s

(3) 5 rad/s

(4) All of these

11) If the rotational kinetic energy of a body is increased by 300% then determine the percentage change in its angular momentum:

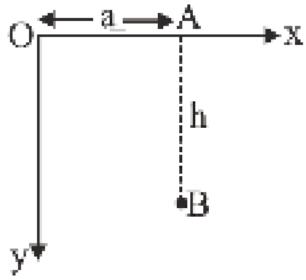
(1) Increases by 100%

(2) Decreases by 100%

(3) Increases by 50%

(4) None of these

12) A particle of mass m is released from rest from point A falling parallel to vertical y-axis. Its



angular momentum after falling through distance h about O is :-

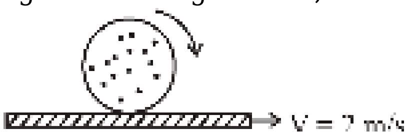
(1) $ma\sqrt{2gh}$

(2) $\frac{m}{a}\sqrt{2gh}$

(3) $\frac{ma}{gh}$

(4) $\frac{mh}{ag}$

13) A uniform disc of radius 20 cm is performing pure rolling on a moving surface, if it's angular



velocity is 5 rad/s then find velocity of it's centre of mass:-

(1) 1 m/s

(2) 2 m/s

(3) 3 m/s

(4) None of these

14) When a solid sphere rolls without slipping down an inclined plane making an angle θ with the horizontal, the acceleration of its centre of mass is a. If the same sphere slides without friction, its acceleration is :-

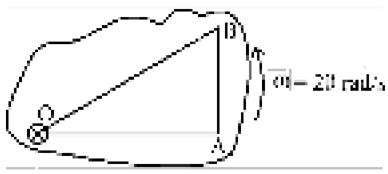
(1) $\frac{7}{2}a$

(2) $\frac{5}{7}a$

(3) $\frac{7}{5}a$

(4) $\frac{5}{2}a$

15) A rigid lamina is rotating about an axis passing perpendicular to its plane through point O as



shown in figure. The angular velocity of point B w.r.t. A is :-

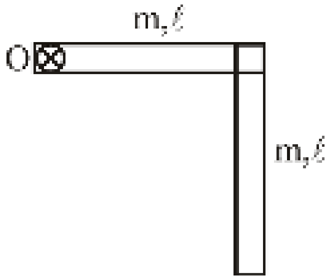
(1) 10 rad/s

(2) 20 rad/s

(3) 40 rad/s

(4) 5 rad/s

16) Find moment of inertia I of the system (Axis is perpendicular to the plane passing through O) :-



(1) $\frac{16}{3}m\ell^2$

(2) $5m\ell^2$

(3) $\frac{2m\ell^2}{3}$

(4) $\frac{5}{3}m\ell^2$

17) A disc of mass M and radius R is reshaped in the form of ring of same mass but radius 2R. The radius of gyration about geometrical axis is increased by a factor :

(1) $\sqrt{2}$

(2) $2\sqrt{2}$

(3) 4

(4) 2

18) A body is rotating with angular acceleration $\alpha = (4t^3 - 3t^2 + 2t)$ starting from rest. Find out angular velocity ω :-

(1) $\omega = t^4 - t^3 + t^2$

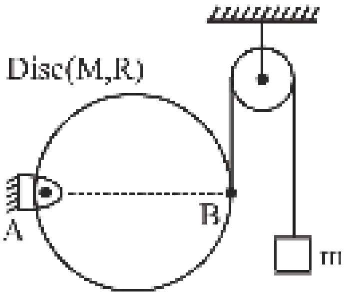
(2) $\omega = t^4 - t^3 + t^2 + 2$

(3) $\omega = 4t^4 - 3t^3 + 2t^2 + 2$

(4) None

19)

Find the value of mass m to keep the disc horizontal. (Given string is attached at point B on disc.)



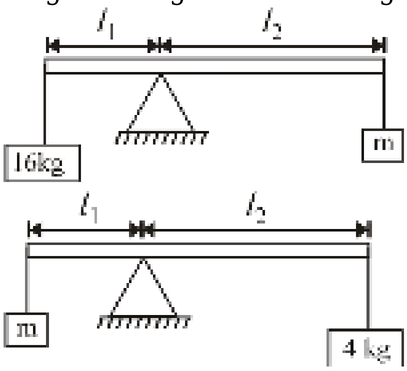
(1) M

(2) $\frac{M}{2}$

(3) $\frac{M}{4}$

(4) $\frac{M}{3}$

20) In an experiment with a beam balance an unknown mass m is balanced by two known masses of 16 kg and 4 kg as shown in figure.



Find the value of the unknown mass m in kg.

(1) 2 kg

(2) 4 kg

(3) 8 kg

(4) 12 kg

21) A disc of mass 2 kg and radius 0.2 m is rotating with angular velocity 30 rad/sec. If a mass of 0.25 kg is put gently on periphery of disc then angular velocity of disc is :-

(1) 24 rad/sec

(2) 36 rad/sec

(3) 15 rad/sec

(4) 26 rad/sec

22) **Assertion :** Kepler's law of areas is equivalent to the law of conservation of angular momentum.

Reason : For planetary motion $\frac{dA}{dt} = \frac{L}{2m}$= constant, where $\frac{dA}{dt}$ is areal speed and L is angular momentum of planet (mass = m) about the sun.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

23) Find gravitational field at point (2, 3) if gravitational potential is given by $V = 5x^2y$.

(1) 0

(2) $-80\hat{i}$

(3) $-60\hat{i}-20\hat{j}$

(4) $-60\hat{i}-80\hat{j}$

24)

Gravitational potential difference between centre and surface of uniform spherical shell of mass M and radius R, is.

(1) $-\frac{GM}{R}$

(2) $-\frac{GM}{2R}$

(3) Zero

(4) $-\frac{3GM}{2R}$

25) The height from earth is surface at which acceleration due to gravity is decreased by 19% of its value at earth surface, is :-

(1) R

(2) $\frac{R}{3}$

(3) $\frac{R}{6}$

(4) $\frac{R}{9}$

26) Two satellites A and B of ratio of masses 3 : 1 are in circular orbit of radius r and 4r. Then ratio of potential energy of A to B is :-

(1) 3 : 1

(2) 1 : 3

(3) 3 : 4

(4) 12 : 1

27) Mass M is divided in two parts aM and (1 – a)M. For a given separation, the value of 'a' for which the gravitational attraction between the two pieces become maximum is :-

(1) 1

(2) $\frac{3}{5}$

(3) 2

(4) $\frac{1}{2}$

28) The Radii of two planets are R and 3R respectively and their densities are ρ and 3ρ. What is the ratio of the acceleration due to gravity at their surface?

(1) 1 : 3

(2) 1 : 1

(3) 1 : 9

(4) 9 : 1

29) Escape velocity of a 1 kg body on the surface of a planet is 100 m/s. Potential energy of this body at the surface of that planet is :-

(1) –5000 J

(2) –1000 J

(3) –2500 J

(4) –10000 J

30) Liquid will rise in a capillary tube to a height of 10 m. If radius of tube become double and arrangement is taken to 50% gravity space then rise of liquid in capillary tube will be :-

(1) 20 m

(2) 40 m

(3) 10 m

(4) None

31) A piece of copper having an internal cavity weights 264 g in air and 221 g in water. Find volume of cavity. Density of Cu = 8.8 g/cc :-

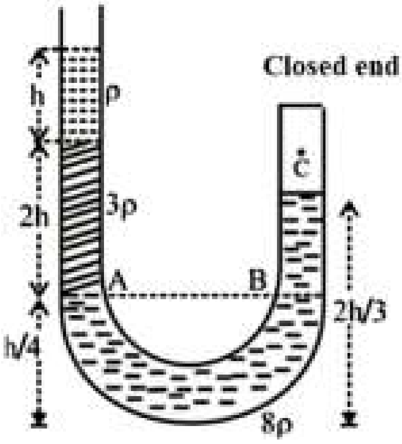
(1) 30 cc

(2) 20 cc

(3) 43 cc

(4) 13 cc

32) A U-tube closed at one end is filled with three immiscible liquids having densities ρ, 3ρ and 8ρ as shown in figure. The air pressure inside the closed end is (P_0 = atmospheric pressure)



(1) $P_0 + \frac{h\rho g}{3}$

(2) $P_0 + \frac{11h\rho g}{3}$

(3) $P_0 + \frac{2h\rho g}{3}$

(4) $P_0 + h\rho g$

33) The neck and bottom of a bottle are 3 cm and 15 cm in radius respectively. If the cork in the neck of bottle is pressed with a force 12 N in the neck of the bottle, then force exerted on the bottom of the bottle is :-

(1) 30 N

(2) 150 N

(3) 300 N

(4) 600 N

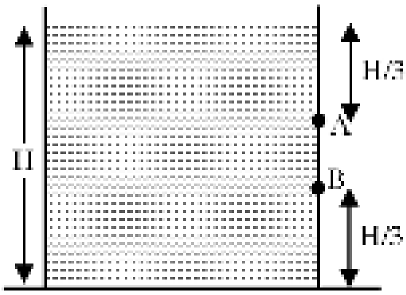
34) A hole is there in the bottom of an open tank having water. If total pressure at bottom is 4 atm (1 atm = 10^5 Pa), then velocity of water flowing from hole (in m/s) is :-

(1) $\sqrt{400}$

(2) $\sqrt{600}$

(3) $\sqrt{100}$

(4) $\sqrt{300}$

35) 

In given figure, two small holes A and B of same area are given. Which of the following is correct ?

(1) Speed of efflux is same for both A & B.

(2) Time of flight is more for A than B.
(3) Range is more for B than A.
(4) Volume flow rate is more at A than B.

36) At what speed, the velocity head of water is equal to pressure head of 50 cm of Hg?

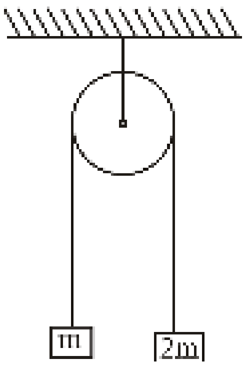
(1) $\sqrt{10}$ m/s
(2) 11.67 m/s
(3) $\sqrt{30}$ m/s
(4) 12.97 m/s

37) A rubber cord catapult has cross sectional area 25 mm^2 and initial length of rubber cord is 10 cm. It is stretched to 15 cm and then released to project a missile of mass 5g. Taking $Y_{rubber} = 5 \times 10^8$ N/m^2. Initial velocity of projected missile is :

(1) 20 ms^{-1}
(2) 100 ms^{-1}
(3) 250 ms^{-1}
(4) 200 ms^{-1}

38) The compressibility of water is 4×10^{-5} per unit atmospheric pressure. The decrease in volume of 100 cc of water under a pressure of 100 atmosphere will be :-

(1) 0.4 cc
(2) 4×10^{-5} cc
(3) 0.025 cc
(4) 0.004 cc

39) Two blocks of masses m and 2m are connected by a metal wire of negligible mass and having cross-sectional area A, passing over a smooth fixed pulley. The masses released from rest than the stress produced in the wire is 

(1) $\frac{mg}{4A}$
(2) $\frac{2mg}{3A}$
(3) $\frac{3mg}{4A}$

(4) $\frac{4mg}{3A}$

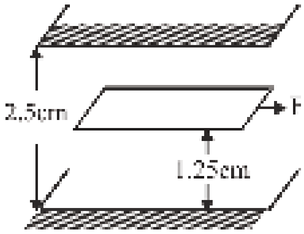
40) The length of a metal wire is ℓ_1 when the tension in it is T_1 and is ℓ_2 when the tension is T_2. The unstretched length of the wire is

(1) $\sqrt{\ell_1 \ell_2}$
(2) $\frac{\ell_1 + \ell_2}{2}$
(3) $\frac{\ell_1 T_2 - \ell_2 T_1}{T_2 - T_1}$
(4) $\frac{\ell_1 T_2 + \ell_2 T_1}{T_2 + T_1}$

41) 64 equal rain drops are falling in air with constant velocity 5 cm/s. If the drops coalesce then the constant velocity of the new drop will be :

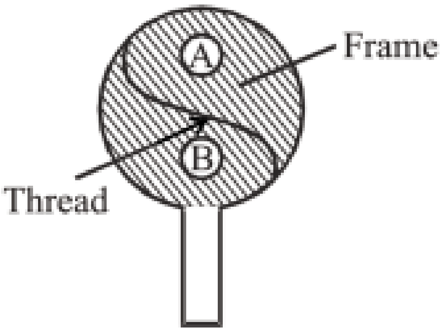
(1) 20 cm/s
(2) 40 cm/s
(3) 60 cm/s
(4) 80 cm/s

42) A space 2.5 cm wide between two large plane surfaces is filled with oil. Force required to drag a very thin plate of area 0.5 m^2 just midway the surfaces at a speed of 0.5 m/sec is 1 N. The coefficient of viscosity in kg-s/m^2 is :



(1) 5×10^{-2}
(2) 2.5×10^{-2}
(3) 1×10^{-2}
(4) 7.5×10^{-2}

43) A thread is tied slightly loose to a wire frame as shown in the figure. And the frame is dipped into a soap solution and taken out. The frame is completely covered with the soap film. When the portion A is punctured with a pin, the thread :



(1) becomes convex towards A

(2) becomes concave towards A

(3) remains in the initial position

(4) either (1) or (2) depending on the size of A w.r.t. B

44) Two spherical soap bubble coalese in air. If V is the consequent change in volume of the contained air and S the change in total surface area then which of the following is correct :
[P = atmospheric pressure, T = Surface Tension]

(1) 4PV + 3ST = 0

(2) 3PV + 4ST = 0

(3) PV + ST = 0

(4) 3PV + ST = 0

45) The material of wire has specific gravity 8. If it does not get wetted by water, what is the maximum diameter (approximately) of the wire that will float on the surface of water ? (T = 70 dyne/cm)

(1) 0.65 mm

(2) 0.65 cm

(3) 1.48 mm

(4) 1.48 cm

CHEMISTRY

1) Which of the following statement is false -

(1) Elements of IB and IIB groups are d-block element

(2) Elements of VA group do not contain metalloids

(3) Elements of IA and IIA groups are normal elements

(4) Elements of IVB group are Ti, Zr, Hf and Rf

2) Which of the following statement/s are not true.
(i) All actinoid elements are not radioactive.
(ii) Radii of noble gases should be compared with covalent radii of other elements.
(iii) Number of electrons in the penultimate shell of d-block elements are 9-18.
(iv) Helium ($1s^2$) does not have completely filled valence shell.

(1) (i), (ii), (iii)

(2) (ii), (iii), (iv)

(3) (i), (ii), (iv)

(4) (i), (iii), (iv)

3)

	Column-I (Electronic configuration)		Column-II (Corresponding elements)
(A)	$[Xe]4f^{14}5d^{2}6s^{2}$	(p)	Inner-transition element
(B)	$[Xe]4f^{14}5d^{10}6s^{2}6p^{6}7s^{2}$	(q)	Representative element
(C)	$[Rn]5f^{14}6d^{1}7s^{2}$	(r)	d-block element
(D)	$[Xe]4f^{14}5d^{10}6s^{2}$	(s)	Transition element
		(t)	s-block element

(1) (A)-r,s; (B) -q,t; (C) -p; (D) -r

(2) (A)-p,r; (B) -p,q,r; (C) -p,s; (D) -t

(3) (A)-q,t; (B) -p; (C) -r; (D) -r,s

(4) (A)-t; (B) -p,r; (C) -p,s; (D) -p,q,r

4) Match list–I and list–II correctly.

	List-I		List-II
1.	Na	P	231 pm
2.	Cl	Q	99 pm
3.	K	R	186 pm
4.	Br	S	114 pm

(1) 1-P, 2-Q, 3-R, 4-S

(2) 1-R, 2-Q, 3-P, 4-S

(3) 1-S, 2-Q, 3-R, 4-P

(4) 1-Q, 2-P, 3-R, 4-S

5) Incorrect order of size is

(1) $H^{-} > Br^{-} > Cl^{-} > F^{-}$

(2) I > Br > Cl > Li

(3) $Mg > Al > Mg^{+2} > Al^{+3}$

(4) $La^{+3} > Ce^{+3} > Gd^{+3} > Lu^{+3}$

6) Match the columns :

	Column -I		Column -II
(A)	Ionization energy	(p)	P > S > Cl

(B)	Atomic radius	(q)	S > Se > Te > O
(C)	Metallic character	(r)	$Be^+ < B^+ < Li^+$
(D)	Electron gain enthalpy	(s)	K > Na > Mg

(1) (A) -p; (B) -r; (C) -s; (D) -q

(2) (A) -r; (B) -p; (C) -s; (D) -q

(3) (A) -r; (B) -q; (C) -p; (D) -s

(4) (A) -s; (B) -r; (C) -p; (D) -q

7) The electronic configuration of some neutral atoms are given below -
(A) $1s^2 2s^1$ (B) $ls^2 2s^2 2p^3$
(C) $1s^2 2s^2 2p^5$ (D) $1s^2 2s^2 2p^6 3s^1$
In which of these electronic configuration would you expect to have highest -
(i) IE_1 (ii) IE_2

(1) C,A

(2) B,A

(3) C,D

(4) B,D

8) Find incorrect statement among the following :-

(1) Electron gain enthalpy is a measure of the ease with which an atom adds an electron to form anion.

(2) Group 17 elements have very high negative electron gain enthalpy.

(3) Noble gases also have large negative electron gain enthalpy.

(4) Generally electron gain enthalpy becomes more negative with increase in atomic number across a period.

9) The increasing order of the first ionization enthalpies of the element B, P, S and F (lowest first) is :-

(1) F < S < P < B

(2) P < S < B < F

(3) B < P < S < F

(4) B < S < P < F

10) Consider the following conversions :
(i) $O(g) + e^- \rightarrow O^-(g)$, ΔH_1
(ii) $F(g) + e^- \rightarrow F^-(g)$, ΔH_2
(iii) $Cl(g) + e^- \rightarrow Cl^-(g)$, ΔH_3
(iv) $O^-(g) + e^- \rightarrow O^{2-}(g)$, ΔH_4
Then according to given information the incorrect statement is:

(1) ΔH_3 is more negative than ΔH_1 and ΔH_2

(2) ΔH_1 is less negative than ΔH_2

(3) ΔH_1, ΔH_2 and ΔH_3 are negative whereas ΔH_4 is positive.

(4) ΔH_1 and ΔH_3 are negative whereas ΔH_2 and ΔH_4 are positive.

11) Arrange the following inert gases in increasing order of positive electron gain enthalpy ($\Delta_{eg}H$)

(1) He < Ne < Ar < Kr < Xe < Rn

(2) Rn < Xe < Kr < Ar < Ne < He

(3) He < Rn < Xe < Ar = Kr < Ne

(4) He < Rn < Xe < Kr < Ar < Ne

12) How many of the following oxides in a aqueous medium turns blue litmus paper into red.
N_2O, P_4O_6, CO, CO_2, Cl_2O_7, NO_2, NO, SO_3.

(1) 5

(2) 8

(3) 4

(4) 7

13) Which of the following is correct ?

(1) HOF > HOCl > HOBr > HOI
(Acidic strength)

(2) $NH_3 > PH_3 > AsH_3 > SbH_3$
(Lewis basic strength order)

(3) HF < HCl < HBr < HI
(Acidic strength)

(4) All of these

14) Which of the following specie has the highest electronegativity ?

(1) C (sp hybridized)

(2) N (sp^2 hybridized)

(3) N (sp hybridized)

(4) C (sp^3 hybridized)

15) How many of the following compound(s) are formed in first excited state of element?
CO_2, BCl_3, XeF_4,SF_6,PCl_5 ?

(1) 2

(2) 5

(3) 4

(4) 3

16) Which is wrong statement?

(1) The decreasing order of bond angle is $H_2O > H_2S > H_2Se > H_2Te$

(2) The decreasing order of bond angle is $NCl_3 > NH_3 > NF_3$

(3) The decreasing order of bond dissociation energy is $Cl_2 > F_2 > Br_2 > I_2$

(4) The decreasing order of boiling point is $H_2O > HF > NH_3$

17) Which of the following statement is incorrect -

(1) The type of hybridisation indicates the geometry of the molecules.

(2) σ bond is formed by parallel overlapping of two p-orbitals.

(3) Strength of bond depends upon the extent of overlapping.

(4) Extent of overlapping in σ bond formation is more than that of π bond so σ bond is stronger than π bond.

18) Which of the following statement is wrong.

(1) All P–Cl bond lengths in PCl_5 are not equal.

(2) Shape of SF_4 is folded square.

(3) $XeOF_4$, PCl_3, SF_4 and NH_3 all have one non bonding electron pair on central atom.

(4) HCl is 100% covalent in nature.

19) Which of the following property is not correct regarding C_2H_2 ?
(a) Diagonal Hybridisation
(b) Contain single super pair
(c) sp - sp overlapping
(d) sp - s overlapping
(e) Contain 5 σ and 2 π bond
(f) 2 lone pairs are present

(1) b, c & d

(2) a, b & c

(3) e & f

(4) c, e & f

20) Which of the following is not correctly matched ?

1.	$ICl_4^{\oplus}$	Folded square shape
2.	XeF_3^+	Bent - T - shape
3.	XeO_6^{4-}	Square pyramidal
4.	PCl_6^-	Square bipyramidal

(1) 1

(2) 2

(3) 3

(4) 4

21) How many of the following molecules having unequal bond length:
$AsCl_5$, IF_7, XeF_4, SF_4, $C\ell F_3$, XeF_2, NO_3^-

(1) 5

(2) 4

(3) 6

(4) 1

22) Which of the following statement is incorrect :-

(1) Both CO_2 and SiO_2 are linear

(2) If internuclear axis is z-axis then p_z orbitals overlaps to form σ bond

(3) The average formal charge on each 'O' atom in ClO_4^- is –1/4.

(4) $2p_x$-$2p_x$ colateral overlapping is stronger than $3p_x$-$3p_x$ colateral overlapping

23) Which molecule is polar but having non polar bond ?

(1) NF_3

(2) PH_3

(3) SO_3

(4) XeF_2

24) Choose incorrect option regarding AB_xL_y -
B = Bonding pair, L = lone pair

(1) x = 5, y = 0 Non-planar as well as non-polar compound.

(2) x = 2, y = 2 Planar as well as polar compound.

(3) x = 2, y = 3 Planar as well as non-polar compound.

(4) x = 7, y = 0 Non-planar as well as polar compound.

25)

Given below are two statements :
Statement-I : The boiling point of hydrides of Group-16 elements follow the order :
$H_2O > H_2Te > H_2Se > H_2S$.

Statement-II : On the basis of molecular mass, H_2O is expected to have lower boiling point than the other members of the group but due to the presence of extensive H-bonding in H_2O, it has higher boiling point.
In the light of the above statements, choose the *correct* answer from the options given below :

(1) Both statement-I and Statement-II are true.

(2) Both statement-I and Statement-II are false.

(3) Statement-I is the true but Statement-II is false.

(4) Statement-I is false but Statement-II is true.

26) Correct match is:-

Column-I		Column-II	
(A)	Chloral hydrate	(p)	Forms Zig-zag chain
(B)	HF (solid)	(q)	Form 2-D-Sheet structure
(C)	$H_3BO_3(s)$	(r)	Inter-molecular H-bond
(D)	H_2SO_4	(s)	Intramolecular H-bond

(1) A–s, B–p, C–q, D–r

(2) A–s, B–q, C–r, D – p

(3) A–p, B–s, C–q, D–r

(4) A–p, B–q, C–r, D–s

27) The solubility of inert gases in water is due to

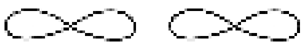
(1) Keesom attraction

(2) Debye attraction

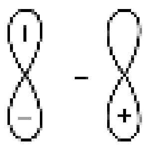
(3) London force

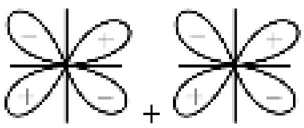
(4) Ion-dipole attraction

28) Which is correctly represent the constructive interference ?

(1) 

(2) 

(3) 

(4) 

29) B_2 molecule will be diamagnetic if -
(I) s - p mixing is not allowed.
(II) s - p mixing is allowed & Hund's rule is violated.
(III) s - p mixing is allowed & Hund's rule is not violated.

(1) I, II

(2) I, III

(3) II, III

(4) I, II & III

30) According to MOT, which has fractional bond order ?

(1) Ne_2
(2) Li_2^+
(3) O_2^{-2}
(4) S_2

31) The correct order of lattice energy of the following compounds

(1) $NaCl > MgCl_2 > CaO > Al_2O_3$
(2) $Al_2O_3 > MgCl_2 > CaO > NaCl$
(3) $MgCl_2 > Al_2O_3 > CaO > NaCl$
(4) $Al_2O_3 > CaO > MgCl_2 > NaCl$

32) Correct order is-

(1) $BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2 > BaCl_2$ (Ionic character)
(2) $AgF > AgCl > AgBr > AgI$ (Covalent character)
(3) $SnCl_2 > SnCl_4$ (Covalent character)
(4) $K_2S > ZnS > CuS$ (Ionic character)

33) How many molecule/species does not exist ?
Be_2, C_2^+, Ne_2, O_2^-, PH_5, BF_6^{3-}, SiF_6^{2-}

(1) 2
(2) 3
(3) 4
(4) 5

34) The number of d-electrons in Fe^{+2} ion is not equal to the number of electron in :-

(1) p-electrons in Ne
(2) s-electrons in Mg
(3) d-electrons in Fe
(4) p-electrons in Cl

35) Supposing the energy of fourth shell for hydrogen atom is –25 a.u (arbitary unit). What would be its potential energy in ground state–

(1) –200

(2) –50

(3) –800

(4) –400

36) One energy difference between the states n = 2 and n = 3 is E eV in hydrogen atom. The ionization potential of H atom is

(1) 3.2 E

(2) 5.6 E

(3) 7.2 E

(4) 13.2 E

37) If the velocity of an electron in Bohr's first orbit is 2.19×10^{6} ms^{-1}, what will be the de Broglie wavelength associated with it ?

(1) 2.19×10^{-6} m

(2) 4.38×10^{-6} m

(3) 3.32×10^{-10} m

(4) 3.32×10^{10} m

38) How many photons of light having a wavelength of 2500Å are necessary to provide 2 joule of energy:-

(1) 1.5×10^{18} photons

(2) 3.5×10^{16} photons

(3) 2.5×10^{18} photons

(4) 5×10^{18} photons

39) The mass of a particle is 10^{-10} g. If its velocity is 10^{-6} cm sec^{-1} with 0.0001% uncertainty in measurement, the uncertainty in its position is :-

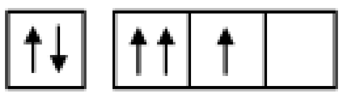
(1) 5.2×10^{-8} m

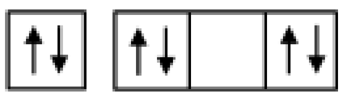
(2) 5.2×10^{-7} m

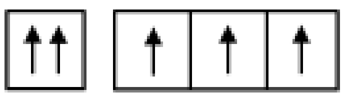
(3) 5.2×10^{-6} m

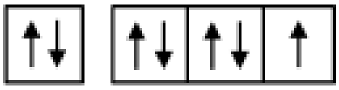
(4) 5.2×10^{-9} m

40) The orbital diagram in which both the Pauli's exclusion principle and Hund's rule are violated is

(1) 

(2) 

(3) 

(4) 

41) Match the column

	Column-I		Column-II
(a)	Nodal plane absent	(p)	$5d_{xy}$
(b)	dumb-bell shaped	(q)	4f
(c)	total nodes three	(r)	2p
(d)	2 angular nodes	(s)	$3d_{z^2}$

(1) a-q, b-p, c-r, d-s

(2) a-q, b-r, c-p, d-s

(3) a-s, b-r, c-q, d-p

(4) a-p, b-q, c-r, d-s

42) If position and momentum of an electron are determined at a time, then ΔP = 3Δx, now the uncertainity in velocity of electron will be :

(1) $\frac{1}{2m}\sqrt{\frac{h}{3\pi}}$

(2) $\frac{1}{2m}\sqrt{\frac{3h}{\pi}}$

(3) $\frac{1}{4m}\sqrt{\frac{3h}{\pi}}$

(4) $\frac{1}{6m}\sqrt{\frac{h}{\pi}}$

43) For sodium atom, the number of electron with m = 0 will be :

(1) 2

(2) 7

(3) 9

(4) 8

44) Orbital angular momentum of 4s electron will be :-

(1) $4\hbar$

(2) 0

(3) $\sqrt{2}\hbar$

(4) $\sqrt{6}\hbar$

45) For Li^{+2} ion, $r_2 : r_5$ will be :-

(1) 9 : 25
(2) 4 : 25
(3) 25 : 4
(4) 25 : 9

BIOLOGY

1) Choose the **incorrect** statement regarding cell wall:-

(1) It provides barrier to all kind of molecules
(2) It gives shape to the cell
(3) It protects the cell from mechanical damage and infection
(4) Secondary cell wall formed on inner side of the primary cell wall

2) Which of the following statement regarding peroxisome is not correct ?

(1) Considered as part of endomembrane system
(2) Bounded by a single membrane
(3) Take part in β-oxidation of fatty acids
(4) In plants they take part in photorespiration

3) Find out the correct match in following :-

	Column-I	**Column-II**	**Column-III**
(i)	Mesosome	Bacteria	Photosynthesis
(ii)	Ribosome	Protein synthesis	Non-membranous
(iii)	Nucleolus	r-RNA-synthesis	Membraneless

(1) i only
(2) i & ii
(3) ii only
(4) ii & iii

4)

Match the following :-

Column-I		**Column-II**	
a	Cristae	I	Flat membranous sacs in stroma
b	Cisternae	II	Infolding in mitochondria
c	Thylakoids	III	Disc shaped sacs in golgi body

(1) a - II; b - III; c - I

(2) a - III; b - II; c - I

(3) a - I; b - II; c - III

(4) a - II; b - I; c - III

5) Given below are two statements :
Statement I : RER is extensive and discontinuous with the outer membrane of the nucleus.
Statement II : A number of proteins synthesised by ribosomes on the endoplasmic reticulum are modified in the cristae of the golgi apparatus before they are released from its trans face.
In the light of the above statement, choose the most appropriate answer from the options given below :-

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct.

6)

Read the following four statements (a-d)
(a) Late prophase → Golgi bodies & ER disappear
(b) Metaphase → Chromosomes arrange on metaphase plate
(c) Anaphase → Centromere split

(d) Telophase → Centriole divide
How many are correctly matched ?

(1) Three

(2) Two

(3) One

(4) Four

7) Which of the following is/are correct for cell division ?
(A) Meiosis I is initiated after the replication of parental chromosomes.
(B) Mitosis leads to restoration of nucleocytoplasmic ratio.
(C) During meiosis, DNA replication and nuclear division occurs twice.

(1) A and B

(2) A and C

(3) A, B and C

(4) Only B

8) Select the correct set of matches:-

(a)	S-phase	DNA replication
(b)	Zygotene	Synapsis

(c)	Pachytene	Dissolution of synaptonemal complex
(d)	Diplotene	Terminalisation of chiasmata

(1) a, b

(2) b, c

(3) c, d

(4) a, d

9) Match the following columns.

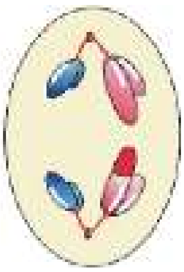
	Column I		**Column II**
(A)	Disintegration of nuclear membrane	(1)	Anaphase
(B)	Reformation of nucleolus	(2)	Prophase
(C)	Division of centromere	(3)	Telophase
(D)	Replication of DNA	(4)	S-phase

(1) A-2, B-3, C-1, D-4

(2) A-2, B-3, C-4, D-1

(3) A-3, B-2, C-1, D-4

(4) A-3, B-2, C-4, D-1

10)  Identify the above stage with its respective event :-

(1) Anaphase II → Splitting of the centromere

(2) Anaphase I → Homologous chromosomes separates

(3) Anaphase I → Splitting of the centromere

(4) Metaphase I → Bivalent chromosomes align on the equatorial plate

11) Read the following statements and choose the incorrect statements.
(a) Palmitic acid has 16 carbons excluding carboxyl carbon.
(b) Lecithin is a phospholipid.
(c) Collagen is the most abundant protein in animal.
(d) Cellulose is a heteropolymer.
(e) Protein is a homopolymer.
Choose the correct options :

(1) (a), (b) and (c)

(2) (c), (d) and (e)
(3) (b), (c) and (d)
(4) (a), (d) and (e)

12) Which of the following statements is correct?

(1) Ricin and morphine are toxins
(2) Concanvalin A is a type of alkaloid
(3) Mono and diterpenes are considered as secondary metabolite
(4) Vinblastin is a type of essentially oil

13) Match the list-I with list-II :

	List-I		**List-II**
(A)	Acidic amino acid	(I)	Valine
(B)	Basic amino acid	(II)	Glutamic acid
(C)	Neutral amino acid	(III)	Phenyl alanine
(D)	Aromatic amino acid	(IV)	Lysine

Choose the correct answer from the options given below :
(1) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)
(2) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
(3) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
(4) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

14) Match the column-I and II :-

	Column-I		**Column-II**
(A)	Substituted methane	(I)	Human haemoglobin
(B)	Secondary metabolite	(II)	Amino acid
(C)	Modified sugar	(III)	Vinblastin
(D)	Quaternary structure	(IV)	Glucosamine

Choose the correct answer from the options given below :-
(1) A-IV, B-III, C-I, D-II
(2) A-II, B-III, C-IV, D-I
(3) A-III, B-I, C-II, D-I
(4) A-I, B-IV, C-II, D-III

15) **Assertion (A) :** In solution of different pH, the structure of Amino acids changes.
Reason (R) : Amino acids have a particular property is the ionizable nature of $—NH_2$ and $—COOH$ groups.
Select the correct option :

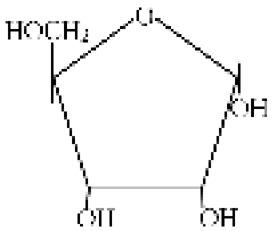
(1) Both Assertion and Reason are correct and `Reason is correct explanation for Assertion.

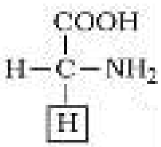
(2) Both Assertion and Reason are correct but Reason is not correct explanation for Assertion.

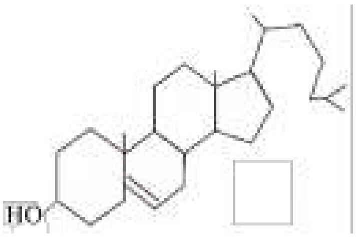
(3) Assertion is true but Reason is false

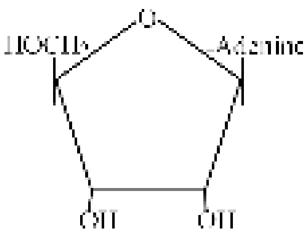
(4) Assertion and Reason both are false

16) Which of the following is micro molecule but insoluble in water ?

(1) 

(2) 

(3) 

(4) 

17) Identify the incorrect statement :-

(1) With increase in substrate concentration the velocity of the enzymatic reaction rises at first.

(2) Enzyme activity is highest at optimum temperature.

(3) Enzymes lower the activation energy of biochemical reaction.

(4) Enzyme are consumed during the reaction they catalyses.

18) Given below are two statements :
Statement-I : Each enzyme has a substrate binding site in its molecule so that a highly stable enzyme-substrate complex is produced.
Statement-II : The binding of the substrate induces the enzyme to alter its shape, fitting more tightly around the substrate.
In the light of above statements, choose the correct answer from the options given below :

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are incorrect.

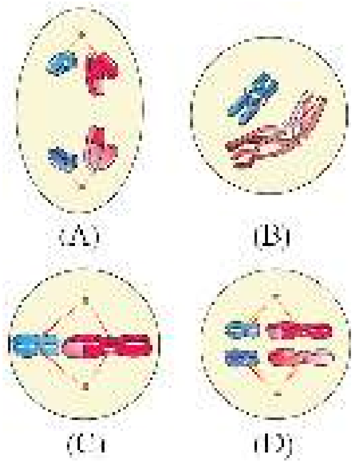
19) **Assertion :-** Lipids are strictly not macro-molecule.
Reason :- Their molecular weight is less than 800 Dalton & form vesicles on grinding.

(1) Both assertion & reason are true & reason is correct explanation of assertion.
(2) Both assertion & reason are true but reason is not correct explanation of assertion.
(3) Assertion is true but reason false.
(4) Both assertion and reason are false.

20) After Meiosis-II, the resultant daughter cells have :-

(1) Same amount of DNA as in the parent cell in S phase
(2) Twice the amount of DNA in comparison to haploid gamete.
(3) Half the amount of DNA in comparison to parent cell
(4) Same amount of DNA as in the parent cell

21) The following figures (A, B, C, D) are showing different stages of meiosis. Choose the option

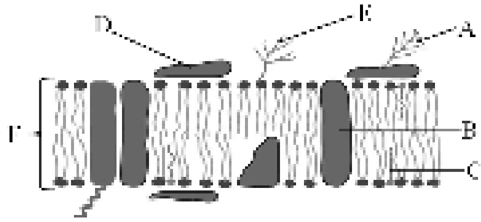


which represents **correct** sequence of given stages.

(1) B → D → A → C
(2) B → D → C → A
(3) A → B → D → C
(4) C → A → D → B

22) How many given below structures commonly found in both cilia and centrioles ?
Radial spokes, Hub, Central sheath, Plasma membrane, Peripheral microtubules, Peripheral triplets, Central microtubules.
Options :-

(1) Four
(2) Three
(3) Two
(4) One

23) 

Which option is wrong according to given labelling.

(1) A – Glycolipid
(2) B – Integral protein
(3) C – Cholesterol
(4) F – Phospholipid bilayer

24) Correctly match column-I, II and III :

	Column-I		**Column-II**		**Column-III**
(A)	Phospholipid	(P)	Uridine	(I)	DNA
(B)	Nucleotide	(Q)	Lecithin	(II)	Codeine
(C)	Nucleoside	(R)	Cytidylic acid	(III)	RNA
(D)	Secondary metabolite	(S)	Alkaloid	(IV)	Cell membrane

(1) A-Q-IV, B-R-III, C-P-II, D-S-I
(2) A-P-I, B-Q-II, C-R-III, D-S-IV
(3) A-Q-IV, B-R-I, C-P-III, D-S-II
(4) A-R-IV, B-Q-I, C-P-II, D-S-III

25) If a protein structure contains 20 Lysine and 10 aspartic acid then the protein will be :-

(1) Acidic
(2) Basic
(3) Neutral
(4) Insoluble

26) In nuclear envelope, what is the role of the perinuclear space ?

(1) Formation of nuclear pore
(2) DNA packaging
(3) Creating a barrier between the nucleus and cytoplasm.
(4) Ribosomal RNA synthesis.

27) In which year Camillo Golgi first observed densely stained reticular structures near the nucleus ?

(1) 1898
(2) 1897
(3) 1887
(4) 1882

28) Which of the following is not common in chloroplasts & mitochondria ?

(1) Both are present in animal cells.
(2) Both contain their own genetic material.
(3) Both are present in eukaryotic cells.
(4) Both are present in plant cells.

29) How many of the following cell organelles are found only in animal cells and not in plant cells ?
(A) Cell wall (B) Centriole (C) Chloroplast (D) Mitochondria (E) 80s ribosome

(1) 1
(2) 2
(3) 3
(4) 4

30) In which phase formation of synaptonemal complex occurs ?

(1) Third stage of prophase-I
(2) Leptotene
(3) Second stage of prophase-I
(4) Diplotene

31) The shape of nerve cell may be-

(1) disc-like
(2) polygonal
(3) columnar
(4) long and branched

32) The osmotic expansion of a cell kept in water is chiefly regulated by –

(1) Mitochondria
(2) Vacuoles
(3) Plastid
(4) Ribosomes

33) In the meiotic cell division four daughter cells are produced by two successive division in which :

(1) First division is reductional and second is equational.
(2) First division is equational, second is reductional.
(3) Both division are equational.
(4) Both division are reductional.

34) Which of the following is true for telophase-I ?

(1) Nuclear membrane reappears

(2) Nucleolus disappears
(3) Chromatin condensed into chromosome
(4) Each chromosome has one chromatid

35) Cytokinesis in animal cell occur in _A_ order and cytokinesis in plant cell occur in _B_ order.

(1) A-Acropetal B-Basipetal
(2) A-Centripetal B-Basipetal
(3) A-Centrifugal B-Centripetal
(4) A-Centripetal B-Centrifugal

36) During prophase, the centrosome which had undergone duplication during ____ phase of interphase, now begins to move towards opposite poles of the cell.

(1) S
(2) G_2
(3) M
(4) G_0

37) In plants, mitotic cell division is seen :

(1) Both in haploid and diploid cells
(2) Only in haploid cell
(3) Only in diploid cell
(4) Only during formation of gamete

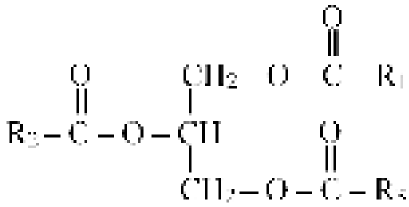
38) In male honey bees mitotic cell division occurs in :-

(1) Haploid cells
(2) Diploid cells
(3) Both haploid and diploid cells
(4) Triploid cells

39)

The four element which constitute upto 95% of all element found in living system are :-

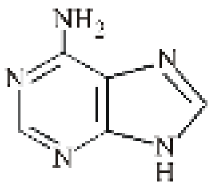
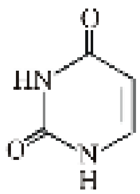
(1) C, H, P, O
(2) C, N, P, O
(3) C, H, O, N
(4) S, O, H, C

40) Identify given structure :- 

(1) Monoglycerides
(2) Diglycerides
(3) Triglycerides
(4) Trihydroxy propane/ Glycerol

41) The acid-insoluble fraction contains organic compounds that have molecular weight the range of ten thousand daltons and above. They include -

(1) Polysaccharide
(2) Amino acids
(3) Nucleic acid
(4) Both (1) and (3)

42) 
(A)

(B)
(A) and (B) in the above diagram are :-

	(A)	(B)
(1)	Guanine	Cytosine
(2)	Adenine	Uracil
(3)	Adenosine	Uridine
(4)	Guanine	Uracil

(1) 1
(2) 2
(3) 3
(4) 4

43) What is variable group in the structure of serine amino acid ?

(1) $–CH_2–NH_2$
(2) $–CH_2–SH$
(3) $–CH_2–OH$
(4) $–CH_2–CH_2–S–CH_3$

44) Which bonds are formed between metal ions and side chains at active site of enzymes ?

(1) Ionic bonds
(2) Covalent bonds
(3) Coordination bonds
(4) Peptide Bonds

45) There could be many more 'altered structural states' between the stable substrate and the product. Implicit in this statement is the fact that all other intermediate structural states are:-

(1) Sometimes stable
(2) Sometimes unstable
(3) Always unstable
(4) Of low energy

46) Mitral valve in mammals guards the opening between :-

(1) Right atrium and Right ventricle
(2) Left atrium and Left ventricle
(3) Right atrium and Left ventricle
(4) Left atrium and Right ventricle

47) Least number of white blood corpuscles are :-

(1) Neutrophils
(2) Basophils
(3) Monocytes
(4) Eosinophils

48)

A mass of nodal tissue called atrio-ventricular node (AVN) is present in :-

(1) Right upper corner of right atrium
(2) Left upper corner of right atrium
(3) Right lower corner of left atrium
(4) Left lower corner of right atrium

49) Select **correct** statement regarding plasma of blood-

(1) Albumin protein primarily involved in defence mechanisms.
(2) High amount of Na^+, Ca^{+2}, HCO_3^-, Cl^- etc present in plasma.
(3) Clotting factor are present in blood in inactive form.
(4) Plasma without the clotting factors is called lymph.

50) Match the column-I with column-II

	Column-I		Column-II

(i)	6-8 % of plasma	(A)	Help in osmotic balance
(ii)	Fibrinogen	(B)	Help in defense mechanism of body
(iii)	Globulin	(C)	Help in coagulation of blood
(iv)	Albumin	(D)	Proteins of plasma

(1) i–D, ii–C, iii–B, IV–A

(2) i–A, ii–D, iii–B, IV–C

(3) i–C, ii–B, iii–A, IV–D

(4) i–B, ii–A, iii–D, IV–C

51) **Assertion :-** A special case of Rh mismatching has been observed between the Rh^- blood of a pregnant mother with Rh^+ blood of the foetus.
Reason :- In case of the subsequent pregnancies, the Rh antibodies from the mother (Rh^-) can leak into the blood of the foetus (Rh^+) & destroy the foetas RBCs and cause erythroblastosis foetalis.

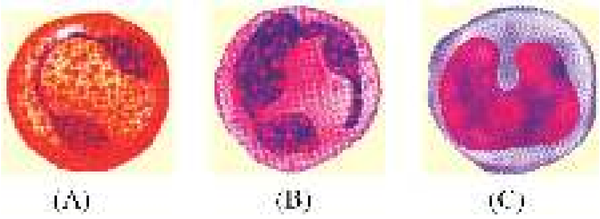
(1) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are correct but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) Both **Assertion** and **Reason** are false.

52) Identify (A,B,C) blood cells in the following figures ? (respectively) :-



	A	B	C
(1)	Monocyte	Neutrophil	Eosinophil
(2)	Neutrophil	Eosinophil	Monocyte
(3)	Eosinophil	Neutrophil	Monocyte
(4)	Neutrophil	Monocyte	Eosinophil

(1) 1

(2) 2

(3) 3

(4) 4

53)

Select the correct order of events occurring in blood clotting :
(I) Formation of fibrin.
(II) Formation of clot.
(III) Thromboplastin formation.
(IV) Conversion of prothrombin to thrombin.

Choose the correct option :

(1) III, IV, II and I
(2) III, II, I and IV
(3) III, IV, I and II
(4) IV, I, III and II

54) Cardiac output increased by :-

(1) Adrenal medullary Hormone
(2) Pitutary gland Hormone
(3) Parathyroid Hormone
(4) Pineal Hormone

55) Time periode of joint diastole is :-

(1) 0.8 sec
(2) 0.4 sec
(3) 0.5 sec
(4) 0.3 sec

56) Read the statements carefully-
(i) If repeated checks of blood pressure of an individual is 140/90 or higher, it shows hypertension.
(ii) High blood pressure leads to heart diseases and also affects vital organs like brain and Kidney.
(iii) A special coronary system of blood vessels is present in our body exclusively for circulation of blood to and from the brain.
(iv) Sequential events in the heart which is cyclically repeated is called the cardiac cycle.
Choose the incorrect statements according to given option-

(1) i & ii only
(2) iv only
(3) i only
(4) iii only

57) Match the column and choose correct option :

	Column-I		**Column-II**		**Column-III**
(A)	Plasma	(P)	Granulocyte	(i)	Immune Response
(B)	Basophil	(Q)	Yellow Colour	(ii)	5.0-5.5 million per mm^3 of blood
(C)	Lymphocyte	(R)	Agranulocyte	(iii)	Histamin Secretion
(D)	Human RBCs	(S)	Enucleated	(iv)	55% of blood

(1) A→ Q, iv ; B→ P, iii ; C→ R, i ; D→ S, ii
(2) A→ Q, iv ; B→ P, iii ; C→ R, ii ; D→ S, i
(3) A→ Q, iv ; B→ R, iii ; C→ P, ii ; D→ S, i
(4) A→ Q, iv ; B→ R, iii ; C→ S, ii ; D→ P, i

58) Time duration between DUP and LUBB sound is:

(1) 0.3 seconds
(2) 0.1 seconds
(3) 0.5 seconds
(4) 0.8 seconds

59) Identify the correct descending order of percentage of leucocyte in human blood ?

(1) Monocyte → eosinophils → Neutrophil → lymphocyte → basophills
(2) Neutrophils → Lymphocytes → Monocyte → eosinophils → basophils
(3) Neutrophils → monocyte → lymphocyte → eosinophils → basophils
(4) Neutrophils → basophils → Lymphocyte → eosinophils → monocyte.

60) Choose incorrect statement :

(1) Per mm^3 of human blood have approx 6000-8000 leucocytes
(2) Neutrophils are most abundant leucocyte in blood
(3) Formed elements of blood constitute nearly 55% of the blood
(4) Reduction in thrombocyte count can lead to excessive loss of blood from body

61) Inter atrial septum in heart of human can be acknowledged as ?

(1) A thin muscular wall
(2) A thick muscular wall
(3) A thin fibrous tissue
(4) A thick fibrous tissue

62) Serum is :

(1) Plasma + clotting proteins
(2) Plasma – clotting proteins
(3) Blood – RBC
(4) Blood – RBC & WBC

63) **Assertion (A) :** Congestion of the lungs is one of the main symptoms of heart failure.
Reason (R) : In heart failure, heart is not pumping blood effectively enough to meet the needs of body.

(1) Both (A) and (R) is true, (R) is not correct explanation of (A)
(2) Both (A) and (R) is true (R) is correct explanation of (A)
(3) Assertion is true but Reason is false
(4) Assertion is false but Reason is true

64) Select the correct statements :-
A. Plasma contains small amounts of minerals like Na^+, Ca^{++}, Mg^{2+}, HCO_3^-, Cl^- etc.

B. Erythrocytes, leucocytes and platelets are collectively called formed elements.
C. Platelets are cell fragments produced from megakaryocytes.
D. Calcium ions play a very important role in clotting.
E. Lymph is also an important carrier for nutrients & hormones.
Options :-

(1) A, B, C and D
(2) B, C and D
(3) A, B, C, D and E
(4) B, D and E

65) About 1200 ml of air is always remain inside the human lungs. It is described as :-

(1) Tidal volume
(2) Residual volume
(3) Expiratory reserve volume
(4) Inspiratory reserve volume

66) Which of the following volume or capacity of lungs can't be measured directly by the spirometer ?

(1) Residual volume
(2) Total lung capacity
(3) Functional residual capacity
(4) All of the above

67) Which has least affinity for hemoglobin ?

(1) Carbonmonoxide
(2) Carbon dioxide
(3) Oxygen
(4) Both (2) & (3)

68) **Assertion :** Expiration takes place when the intrapulmonary pressure is higher than the atmospheric pressure.
Reason : Relaxation of the diaphragm and the intercostal muscles returns the diaphragm and sternum to their normal positions and reduce the thoracic volume and thereby the pulmonary volume.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

69)

(i) Increased intrapulmonary pressure

(ii) Decreased thoracic cavity volume
(iii) Contraction of Diaphragm and EICM
(iv) Increased thoracic cavity volume
(v) Relaxation of diaphragm and EICM

How much of the above events are related with initiation of normal inspiration

(1) Five
(2) Four
(3) Three
(4) Two

70) Which one of the following statements is wrong ?

(1) Solubility of CO_2 is 20–25 times higher than that of O_2
(2) 70% CO_2 is transported by plasma as bicarbonate
(3) Occupational respiratory disorders are related to stone breaking industries.
(4) Chemosensitive area is located above pons

71) Which respiratory capacity is correctly matched with its formula & value :-

	Respiratory capacities	**Formula**	**Value**
(1)	Functional residual capacity	ERV + RV	2300 ml
(2)	Vital capacity	ERV+RV+IRV	4600 ml
(3)	Total lung capacity	RV+IRV+ERV	5800 ml
(4)	Inspiratory capacity	TV+IRV	1600 ml

(1) 1
(2) 2
(3) 3
(4) 4

72)

Match the following columns :-

Respiratory organ		**Animal**	
A	Moist skin	Q	Scorpion
B	Tracheal Tubes	R	Limulus
C	Gills	S	Frog
D	Book gill	T	Insect
E	Book lung	U	Prawn

(1) A-Q, B-R, C-S, D-T, E-U
(2) A-S, B-T, C-U, D-R, E-Q

(3) A-U, B-T, C-S, D-R, E-Q

(4) D-Q, B-T, C-U, E-S, A-R

73) **Statement-I :-** Every 100 mL of deoxygenated blood delivers approximately 4 mL of CO_2 to the alveoli.
Statement-II :- Our lungs remove large amount of CO_2 every day.

(1) Statement-I and statement-II both are true.

(2) Statement-I is true and statement-II is false.

(3) Statement-I is false and statement-II is true.

(4) Statement-I and statement-II both are false.

74) Besides RBC, blood plasma also carries O_2 in soluble state. The % is :-

(1) 1%

(2) 7%

(3) 3%

(4) 3–6%

75) Oxygen dissociation curve shifts to right when :-

(1) pO_2 increases

(2) pCO_2 decreases

(3) pH decreases

(4) Temperature decreases

76) Which of the following factors is not favourable for gaseous diffusion across diffusion membrane :-

(1) High diffusion capacity

(2) High solubility

(3) Thickness of diffusion membrane is more than one millimetre

(4) Higher difference in partial pressure of gases

77) Some statements are given, read them and choose disorder :-
(i) Allergic condition
(ii) Difficulty in breathing
(iii) Increased I_gE
(iv) Wheezing sound

(1) Emphysema

(2) Anthracis

(3) Silicosis

(4) Asthma

78) Which of the following substances are reabsorbed actively in nephron ?

(1) Glucose, Ammonia

(2) Glucose, Na^+

(3) Amino acid, Urea

(4) Na^+, Uric acid

79) Which of the following group of animals are ureotelic in nature?

(1) Human, tadpole of frog

(2) Human, frog

(3) Crustaceans, frog

(4) Spider, earthworm

80) Which of nitrogenous waste also eliminated through saliva ?

(1) Urea

(2) Creatine

(3) HCO_3^-

(4) TMAO

81) **Assertion :** The ascending limb is impermeable to water but allows transport of electrolytes actively or passively.
Reason : In ascending limb filtrate gets concentrated due to the passage of electrolytes into the medullary fluid.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

82) Match the following:-

Column - I		**Column - II**	
(A)	Cockroach	(i)	Protonephridia
(B)	Earthworm	(ii)	Malpighian tubules
(C)	Prawn	(iii)	Nephridia
(D)	Amphioxus	(iv)	Antennal glands

(1) A – ii, B – iii, C – i, D – iv

(2) A – iii, B – ii, C – i, D – iv

(3) A – iv, B – iii, C – ii, D – i

(4) A – ii, B – iii, C – iv, D – i

83) Suppose you are developing a new drug, and have found that when it is administered in humans there is a substantial increase in the volume of urine produced. When you administer antidiuretic hormone (ADH or vasopressin) at the same time, the volume of urine returns to normal. Which

hypothesis best fits these observations? The new drug?

(1) Blocks the receptors for ADH on the collecting duct of the kidney.
(2) Blocks the release of ADH from the pituitary
(3) Mimics the action of ADH
(4) Decreases blood pressure

84) Read the given statements :
(A) On inner convex surface of kidney a notch is present called hilum
(B) The cortex extends in between the medullary pyramids as renal column called columns of Bertini
(C) A rise in glomerular blood pressure can activate the JG cells to release renin.
(D) Sweat produced by the sweat glands is a watery fluid containing NaCl, small amounts of urea, lactic acids etc.
Select the correct answer :

(1) B and C are correct
(2) A and C are incorrect
(3) C and D are correct
(4) A and D are incorrect

85) **Statement-I :** In aquatic amphibians and aquatic insects ammonia is excreted by kidneys. Kidneys play significant role in its removal.
Statement-II : Flame cells are the tubular excretory structures of earthworm and other annelids.
In the light of above statements, choose the correct answer from the options given below.

(1) Statement-I is correct but statement-II is incorrect.
(2) Statement-I is incorrect but statement-II is correct.
(3) Both Statement-I and statement-II are correct.
(4) Both Statement-I and statement-II are incorrect.

86) Which of the following statement is not correct about kidney ?

(1) Kidneys are reddish brown, bean shaped structures.
(2) Kidneys situated between last thoracic and fourth lumbar vertebra.
(3) Towards the centre of inner concave surface of the kidney is a notch called hilum.
(4) Inside the kidney there are two zones, an outer cortex and inner medulla.

87) Effective filtration pressure in glomerulus is caused due to :-

(1) Powerful pumping action of the heart
(2) Secretion of adrenaline
(3) Afferent arteriole is slightly wider than efferent arteriole
(4) Afferent arteriole is slightly narrower than efferent arteriole

88) What happens if there is excessive loss of body fluid from the body ?

(1) Excessive loss of body fluid can switch off osmoreceptors

(2) This suppress the ADH release from hypothalamus

(3) Stimulate the hypothalamus to release vasopressin by activating the osmoreceptors

(4) ADH facilitates water reabsorption from PCT

89) Urine formation involve various steps, out of them which step is counted as a non selective process ?

(1) Ultrafiltration

(2) Reabsorption

(3) Secretion

(4) Both (2) and (3)

90) Read the following statements-
(A) The glomerular capillary blood pressure causes filtration of blood through 3 layers.
(B) GFR in a healthy individual is approximately 1100-1200 ml/minute.
(C) Reabsorption of water occurs actively in the initial segments of the nephron.
(D) Tubular secretion helps in the maintenance of ionic and acid base balance of body fluids.
How many of the above statements are correct?

(1) Three

(2) Two

(3) Only One

(4) Four

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**3**	**2**	**2**	**1**	**3**	**3**	**4**	**4**	**3**	**4**	**1**	**1**	**3**	**3**	**2**	**4**	**2**	**1**	**2**	**3**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**1**	**1**	**3**	**3**	**4**	**4**	**4**	**3**	**1**	**3**	**4**	**2**	**3**	**2**	**2**	**2**	**3**	**1**	**4**	**3**
Q.	41	42	43	44	45															
A.	**4**	**2**	**2**	**2**	**3**															

CHEMISTRY

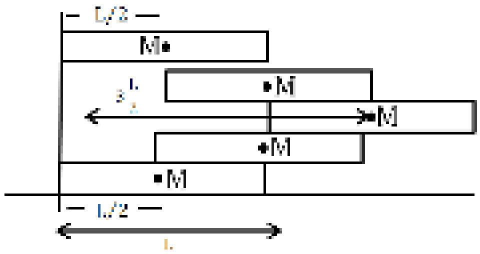
Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	**2**	**3**	**1**	**2**	**2**	**2**	**1**	**3**	**4**	**4**	**3**	**1**	**4**	**3**	**4**	**3**	**2**	**4**	**3**	**3**
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	**2**	**1**	**2**	**4**	**1**	**1**	**2**	**3**	**1**	**2**	**4**	**4**	**3**	**4**	**3**	**3**	**3**	**3**	**1**	**1**
Q.	86	87	88	89	90															
A.	**3**	**2**	**2**	**2**	**2**															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	**1**	**1**	**4**	**1**	**2**	**1**	**1**	**1**	**1**	**1**	**4**	**3**	**1**	**2**	**1**	**3**	**4**	**3**	**1**	**3**
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	**1**	**3**	**1**	**3**	**2**	**3**	**1**	**1**	**1**	**3**	**4**	**2**	**1**	**1**	**4**	**1**	**1**	**1**	**3**	**3**
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	**4**	**2**	**3**	**3**	**3**	**2**	**2**	**4**	**3**	**1**	**1**	**3**	**3**	**1**	**2**	**4**	**1**	**3**	**2**	**3**
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	**1**	**2**	**2**	**3**	**2**	**4**	**2**	**4**	**4**	**4**	**1**	**2**	**1**	**3**	**3**	**3**	**4**	**2**	**2**	**1**
Q.	171	172	173	174	175	176	177	178	179	180										
A.	**3**	**4**	**2**	**2**	**4**	**2**	**3**	**3**	**1**	**2**										

SOLUTIONS

PHYSICS

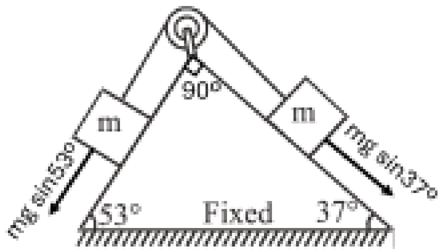


1)

$$X_{com} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

$$X_{com} = \frac{M \times \frac{L}{2} + M \times L + M \times \frac{3L}{2} + M \times L + M \times \frac{L}{2}}{M + M + M + M + M}$$

$$X_{com} = \frac{9L}{10}$$



2)

Acceleration of blocks (along the incline plane)

$$a = \frac{mg(\sin 53^\circ - \sin 37^\circ)}{2m}$$

$$a = \frac{g}{2}(\sin 53^\circ - \sin 37^\circ)$$

$$a = \frac{10\ m/s^2}{2}(0.8 - 0.6)$$

$$a = 5\ m/s^2(0.2) = 1\ m/s^2$$

$$\vec{a}_{com} = \frac{m\vec{a}_1 + m\vec{a}_2}{m + m} = \frac{\vec{a}_1 + \vec{a}_2}{2}$$

$\vec{a}_1$ and $\vec{a}_2$ **are perpendicular to each other, so**

$$|\vec{a}_{com}| = \frac{1}{2}\sqrt{a_1^2 + a_2^2} = \frac{1}{2}\sqrt{a^2 + a^2}$$

$$= \frac{1}{\sqrt{2}}a = \frac{1}{\sqrt{2}}(1) = \frac{1}{\sqrt{2}} m/s^2$$

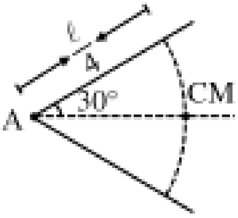
3) $\left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 = \left(\frac{m - 4m}{m + 4m}\right)^2 = \frac{9}{25}$

4) For perfectly elastic collision , e = 1, $(KE)_{BC} = (KE)_{AC}$

For inelastic collision 0 < e < 1, KE is lost.

For perfectly inelastic collision, e = 0, KE is lost momentum is conserved in all types of collision, in case of bomb explosion, momentum is conserved and kinetic energy increases.

5) Distance $= \frac{\ell}{4}\cos 30 = \frac{\ell\sqrt{3}}{8}$



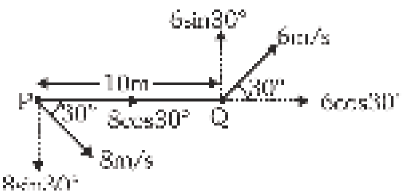
6) Total acceleration $a = \sqrt{a_c^2 + a_t^2}$

$5 = \sqrt{\left(\frac{v^2}{r}\right)^2 + 3^2} = \sqrt{\left(\frac{12^2}{r}\right)^2 + 3^2}$

$\Rightarrow r = 36m$

7) $h = \frac{5}{2}r \Rightarrow r = \frac{2}{5} \times h = \frac{2}{5} \times 5 = 2$ metre

8)

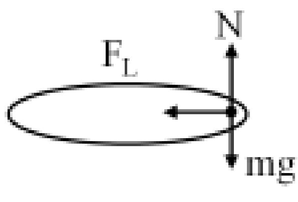


$\omega_{PQ} = \frac{(V_{PQ})_{\perp}}{r} = \frac{8\sin 30° + 6\sin 30°}{10}$

$\omega_{PQ} = 0.7$ rad/s

9) $\mu = \frac{rg}{v^2} = \frac{36 \times 10}{900} = 0.4$

10)



$F_L = \mu N$

$\mu N \geq m\omega^2 r$

$\mu mg \geq m\omega^2 r$

$\omega^2 r \leq \mu g$, $\omega_{max} = \sqrt{\frac{\mu g}{r}}$

$\omega_{max} = \sqrt{\frac{1}{2}\frac{9.8}{(0.1)}} = 7$ rad/sec

11)

$KE \propto L^2$

$$\frac{\Delta L}{L} = \left[\sqrt{\frac{KE_F}{KE_i}} - 1\right] \times 100$$

$$\frac{\Delta L}{L} = \left[\sqrt{\frac{400}{100}} - 1\right] \times 100$$

$$\frac{\Delta L}{L} = 100\%$$

12) $L = r\,p$

$v = \sqrt{2gh}$

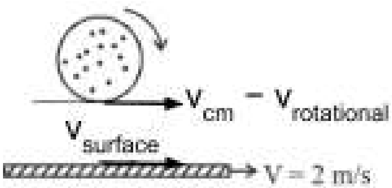
Using the equation of motion for free fall:

$v^2 = 0^2 + 2gh$

$v = \sqrt{2gh}$

Angular momentum = (momentum) (perpendicular distance)

= (mv) (a) = $ma\sqrt{2gh}$

13) 

The condition for pure rolling is that the velocity of the point of contact on the rolling body is equal to the velocity of the surface it is rolling on. The velocity of the point of contact is the vector sum of the translational velocity of the center of mass and the rotational velocity of that point.

$v_{cm} - v_{rotational} = v_{surface}$

$v_{cm} = v_{surface} + v_{rotational}$

$v_{rotational} = \omega R$

$v_{cm} = v_{surface} + \omega R$

Radius of the disc, R = 20 cm = 0.20 m

Angular velocity of the disc, ω = 5 rad/s

Velocity of the moving surface, vsurface = 2 m/s

$v_{cm} = 2 + (5)(0.20)$

$v_{cm} = 2 + 1.0 = 3$ m/s

14)

$a_{sliding} = g\sin\theta$

$$a_{rolling} = \frac{g\sin\theta}{1 + \frac{2}{5}}$$

$$a_{rolling} = (g\sin\theta)\left(\frac{5}{7}\right)$$

$$a_{sliding} = \frac{7}{5}\, a_{rolling}$$

15)

The relative angular velocity between two points on a rigid body is the same as the angular velocity of the rigid body.

16)

$I = I_1 + I_2$

$$I = \frac{ml^2}{3} + \left(\frac{ml^2}{12} + mx^2\right)$$

where $x^2 = l^2 + l^2/4$

$$I = \frac{ml^2}{3} + \left(\frac{ml^2}{12} + m\left(l^2 + \frac{l^2}{4}\right)\right)$$

$$I = \frac{5}{3}ml^2$$

17) **Solution:** Moment of inertia of the disc of mass M and radius R

$I_{disc} = Mk_1^2 = \frac{1}{2}MR^2 \Rightarrow k_1 = \frac{R}{\sqrt{2}}$ And for the ring of same mass but radius 2R,

$I_{ring} = Mk_2^2 = M(2R)^2 \Rightarrow k_2 = 2R$

Thus, $\frac{k_2}{k_1} = 2\sqrt{2}$

18) **Calculation**

Given: $\alpha(t) = 4t^3 - 3t^2 + 2t$

$$\omega(t) = \int \left(4t^3 - 3t^2 + 2t\right) dt = \left(\frac{4t^4}{4} - \frac{3t^3}{3} + \frac{2t^2}{2}\right) + C$$

$$= t^4 - t^3 + t^2 + C$$

Initial condition: at $t = 0$, $\omega = 0$

$0 = 0 - 0 + 0 + C \Rightarrow C = 0$

$\omega(t) = t^4 - t^3 + t^2$

19)

To keep the disc horizontal

Net torque = 0 (about point A)

$Mg(R) - T(2R) = 0$

$MgR - mg(2R) = 0$ $(T = mg)$

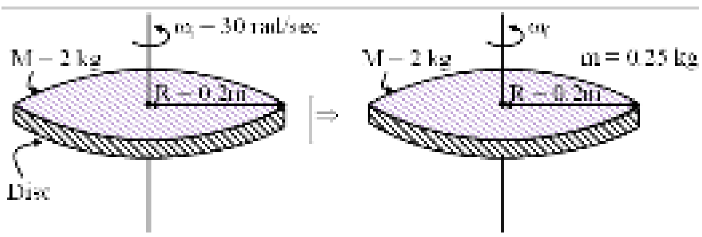
$$\boxed{m = \frac{M}{2}}$$

20) **Question Explanation :** In the experiment the beam balance is in the state of equilibrium in two ways as shown. We need to find the unknown mass m.

Concept : Equilibrium of rigid bodies.

Solution :

$mg\,\ell_2 = 16\,g\ell_1$ .....(i)
$mg\,\ell_1 = 4g\ell_2$ .....(ii)
$\Rightarrow \frac{16}{m} = \frac{m}{4} \Rightarrow m = 8$ kg

21) 

Solution :
L_{system} = constant
$I\omega$ = constant
$I_i\omega_i = I_f\omega_f$
$$\left(\frac{MR^2}{2}\right)(\omega_i) = \left(\frac{MR^2}{2} + mR^2\right)(\omega_f)$$
$$\left[\frac{2 \times (0.2)^2}{2}\right][30] = \left[\frac{2 \times (0.2)^2}{2} + 0.25(0.2)^2\right][\omega_f]$$
$0.04 \times 30 = (0.04 + \frac{1}{4} \times 0.04)\,\omega_f$
$1.2 = (0.4 + 0.01)\omega_f$
$1.2 = (0.05)\omega_f$
$\frac{1.2}{0.05} = \omega_f$
$\omega_f = \frac{120}{5}$
$\omega_f = 24$ rad/sec

22)

Kepler's law of area states that the areal velocity dA/dt of a planet about the Sun is constant.

For planetary motion, angular momentum $L = m\,r^2\omega$, and areal speed $\frac{dA}{dt} = \frac{1}{2}r^2\omega$.
Thus $\frac{dA}{dt} = \frac{L}{2m}$, which is constant when angular momentum is conserved.
Hence, Kepler's law of areas is equivalent to the law of conservation of angular momentum.

23)

Given $V = 5x^2y$
$E_x = \frac{-\partial V}{\partial x} = -10xy$; $E_y = \frac{-\partial V}{\partial y} = -5x^2$
at point (2, 3) ; $\vec{E} = E_x\hat{i} + E_y\hat{j} = -60\hat{i} - 20\hat{j}$

24) For Shell → potential is uniform inside the uniform shell

$\Delta V = 0$.

25) $g_h = \dfrac{g}{\left(1+\frac{h}{R}\right)^2}$

$\dfrac{81}{100}g = \dfrac{g}{\left(1+\frac{h}{R}\right)^2}$

$\dfrac{9}{10} = \dfrac{1}{\left(1+\frac{h}{R}\right)}$

$9 + \dfrac{9h}{R} = 10$

$\dfrac{9h}{R} = 1$

$h = R/9$

26) Potential energy of a satellite = $\dfrac{-GMm}{r}$

$\dfrac{PE_A}{PE_B} = \dfrac{m_A}{m_B} \times \dfrac{r_B}{r_A} = \dfrac{3}{1} \times \dfrac{4}{1} = \dfrac{12}{1}$

27) $F_g \propto$ (Product of masses)

$\propto aM \times (1-a)M \propto aM^2(1-a)$

For maximum force, $\dfrac{dF}{da} = 0$

$\Rightarrow M^2 - 2aM^2 = 0$

$a = \dfrac{1}{2}$

28) $g = \dfrac{4}{3}\pi R\rho G$

$g \propto R\rho$

$\dfrac{g_1}{g_2} = \dfrac{R_1}{R_2} \times \dfrac{\rho_1}{\rho_2} = \dfrac{R}{3R} \times \dfrac{\rho}{3\rho}$

$\dfrac{g_1}{g_2} = \dfrac{1}{9}$

29) $\sqrt{\dfrac{2GM}{R}} = 100$

$\dfrac{GM}{R} = \dfrac{10000}{2} = 5000$

$U = -\dfrac{GMm}{R} = -5000 \times 1$

$U = -5000$ J

30) $h = \dfrac{2T\cos\theta}{\rho g r}$

$g_2 = \dfrac{g}{2}$, $r_2 = 2r$, $h_2 = h = 10$ m

31)

Thrust $Th = V\sigma g$

Vol. of body $V = \dfrac{(264-221)g}{1\times g} = 43cc$

volume of material $= \dfrac{\text{mass of body}}{\text{density}} = \dfrac{264}{8.8} = 30\,cc$

vol. of cavity = (43 – 30) cc = 13 cc

32) $P_A = P_B$ $P_0 + h\rho g + (2h)(3\rho)g = P_{tube} + \left(\dfrac{2h}{3} - \dfrac{h}{4}\right)(8\rho)\,g$

$P_0 + 7h\rho g = P_{tube} + \dfrac{10}{3}h\rho g$ $\Rightarrow$ $P_{tube} = P^0 + \dfrac{11}{3}h\rho g$

33) $\dfrac{F_1}{A_1} = \dfrac{F_2}{A_2}$

$F_2 = \left(\dfrac{A_2}{A_1}\right) F_1 = \left(\dfrac{R_2}{R_1}\right)^2 F_1$

$= \left(\dfrac{15}{3}\right)^2 \times 12 = 300\,N$

34)

$P_{Bottom} = P_0 + \rho gH$

4 atm = 1 atm + ρgH

ρgH = 3 atm = 3×10^5 Pa

H = 30 m

$v = \sqrt{2gH} = \sqrt{600}$ m/s

35) $t = \sqrt{\dfrac{2h}{g}}$ $(t_A > t_B)$

36) $\dfrac{V^2}{2g} = \dfrac{P_{Hg}}{\rho_{wg}} = \dfrac{\rho_{Hg}\times g\times h_w}{\rho_w \times g}$

$\dfrac{v^2}{2g} = 0.5 \times 13.6$

$v^2 = 136$

v = 11.67 m/s

37) By energy conservation

$$\frac{1}{2}F\,\Delta\ell = \frac{1}{2}mv^2$$

$$\frac{YA(\Delta\ell)^2}{\ell} = mv^2$$

38) $B = \frac{1}{C} = -\frac{\Delta P}{\Delta V/V}$

$\Rightarrow \quad \Delta V = -(\Delta P)VC$

$|\Delta V| = 100 \times 100 \times 4 \times 10^{-5}$

$= 0.4$ cc

39) $a = \frac{2mg - mg}{3m} = \frac{g}{3}$

$T = m(g + a)$

$$T = m\left(g + \frac{g}{3}\right) = \frac{4mg}{3}$$

stress $= \frac{T}{A} = \frac{4mg}{3A}$

40) $\Delta\ell = \frac{TL}{AY}$

$T \propto \Delta\ell$

$$\frac{T_1}{T_2} = \frac{\ell_1 - L}{\ell_2 - L}$$

$T_2\ell_1 - T_2L = T_1\ell_2 - T_1L$

$$L = \frac{T_2\ell_1 - T_1\ell_2}{T_2 - T_1}$$

41) $64\left(\frac{4}{3}\pi r^3\right) = \frac{4}{3}\pi R^3$

$4r = R$

$$\frac{v_1}{v_2} = \left(\frac{r}{R}\right)^2$$

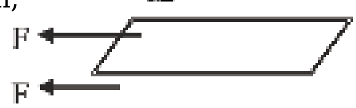
$$\frac{5}{v_2} = \left(\frac{1}{4}\right)^2$$

$v_2 = 80$ cm/s

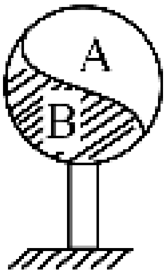
42) Velocity gradient $= \dfrac{0.5}{\frac{2.5}{2} \times 10^{-2}}$

as force on the plate due to viscosity is from upper as well as lower portion of the oil, equal from each part,

Then, $F = 2\eta A\frac{dv}{dz} = 2 \times \eta \times (0.5)\,\dfrac{0.5}{1.25 \times 10^{-2}}$



$\Rightarrow \eta = 2.5 \times 10^{-2}$ kg – sec/m^2

43) 

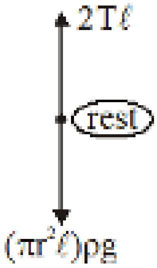
44) In air $P_1V_1 + P_2V_2 = PV$

$$\left(P_0 + \frac{4T}{r_1}\right)\frac{4}{3}\pi r_1^3 + \left(P_0 + \frac{4T}{r_2}\right)\frac{4}{3}\pi r_2^3 = \left(P_0 + \frac{4T}{r}\right)\frac{4}{3}\pi r^3$$

$$P_0\left(\frac{4}{3}\pi r_1^3 + \frac{4}{3}\pi r_2^3 - \frac{4}{3}\pi r^3\right)$$

$$+\frac{4T}{3}(4\pi r_1^2 + 4\pi r_2^2 - 4\pi r^2) = 0$$

$$\Rightarrow P_0V + \frac{4T}{3}S = 0$$

45) 

for max diameter $\boxed{\cos\theta = 1}$

$2T\ell = \pi r^2 \ell \rho g$

$$r = \sqrt{\frac{2T}{\pi\rho g}} = \sqrt{\frac{2 \times 0.07}{\pi \times 8 \times 10^4}} = \sqrt{\frac{7 \times 10^{-6}}{12.56}}$$

$= 0.74$ mm

$d = 2r = 1.48$ mm

CHEMISTRY

46)

NCERT-XI, Pg. # 77 to 79

47)

NCERT-XI, Pg. # 79, 81

48)

NCERT-XI, Pg. # 77, 78, 79

49) NCERT date 3.4 pages 87

50)

Cl < Br < I < Li

Covalent Radius	Metallic Radius	(Metallic Radius	> Covalent Radius)

51)

NCERT-XI, Pg. # 81-86

52) **Explanation & Solution :** For IE1 – option C would have highest first IP because of high $Z_{effective}$ among all. It is configuration of fluorine which has highest 1st IP in 2nd period after Neon
For IE2 — option 1 will have highest 2nd IP because after removing one electron from 1 we will get configuration of helium which has highest IP in periodic table so 2nd IP in option 1 will be maximum.
Answer option (1)

53)

Solution : Noble gases have stable, filled electron shells. Adding another electron is energetically unfavorable, so their electron gain enthalpy is positive, not large and negative.
Final answer : 3

54) Element : B S P F
I.P.(kJ mol^{-1}) : 801 1000 1011 1681
In general as we move from left to right in a period, the ionization enthalpy increases with increasing atomic number. The ionization enthalpy decreases as we move down a group. P($1s^2$, $2s^2$, $3s^2$ $3p^3$) has stable half filled electronic configuration than S($1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^4$). For this reason, ionization enthalpy of P is higher than S.

55)

First electron gain enthalpies are usually negative (exothermic). Second electron gain enthalpies are positive (endothermic) due to electron repulsion. Electron gain enthalpy becomes more negative across a period (F > Cl > O).
ΔH_2 (F) is more negative than ΔH_1.
ΔH_3 (Cl) is more negative than ΔH_1 (O) and ΔH_2 (F).
ΔH_1 (O) is less negative than ΔH_2 (F) whereas ΔH_4 (O^- to O^{2-}) is positive.

56) ($\Delta_{eg}H$ value)
He = +48, Ne = +116, Ar = +96, Kr = +96, Xe = + 77, Rn = +68

57) NCERT, Pg. # 91

58)

As per concept

59)

Solution : Electronegativity increases with 's' character (sp > sp^2 > sp^3). Nitrogen is also inherently more
electronegative than Carbon. Therefore, an sp-hybridized Nitrogen is the most electronegative.
% s-character ∝ electronegativity.
Final answer : 3

60)

$\left.\begin{array}{l} CO_2 \\ BCl_3 \\ PCl_5 \end{array}\right] \text{I E.S.}$

$XeF_4 \Rightarrow$ II E.S.

$SF_6 \Rightarrow$ II E.S.

E.S. → Excited state

61)

B.E ⇒ $Cl_2 > Br_2 > F_2 > I_2$

62) NCERT Pg.# 119, 120

63)

No bond is 100% ionic or 100% covalent.

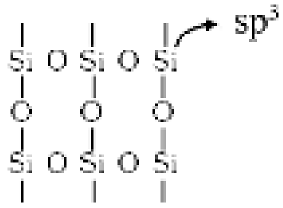
64) H – C ≡ C – H
sp hybridisation

65) NCERT, Pg. # 111 (Table - 4.6)

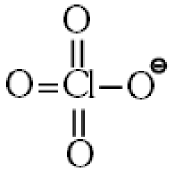
66)

Answer: 4

67) (i) O = C = O (linear)



(Tetrahedral)

(iii)  average f.c. on oxygen = $-\frac{1}{4}$

68)

P-H bond non-polar PH_3 molecule is polar

69)

Solution:
VSEPR: total electron pairs = x + y → electron-pair geometry → molecular shape. Polarity: symmetric shapes with identical ligands are generally non-polar (dipoles cancel).
1. x = 5, y = 0 → AX_5 = trigonal bipyramidal (e.g., PCl_5). Non-planar, dipoles cancel → non-polar.
2. x = 2, y = 2 → AX_2E_2 = bent (H_2O type). Planar and polar.
3. x = 2, y = 3 → AX_2E_3 = linear (XeF_2). Planar and non-polar.
4. x = 7, y = 0 → AX_7 = pentagonal bipyramidal (IF_7). Non-planar but with identical ligands it’s non-polar, not polar.
Final Answer: (4)

70)

Boiling point order: $H_2O > H_2Te > H_2Se > H_2S$ — correct (Statement I true).
Reason: H_2O forms strong H-bonds, giving higher b.p. than expected from molecular mass — also true.
Final Answer: Option (1)

71)

$CCl_3CH(OH)_2$ = Chloral Hydrate
(Intra molecular H-Bond)
H_2SO_4 = inter molecular H-Bond
HF (s) = Zig-Zag Chain structure
H_3BO_3(s) = 2D Sheet

72) Inert gas + H_2O
⇒ Dipole-induced dipole

(Debye attraction)

73)

NCERT-XII, Pg. # 115

74) s-p mixing is applicable till N_2 due to which there is energy change b/w σ2pz & π2px = π2py

75)

(1) Ne_2 B.O = 0
(2) Li_2^+ B.O = 0.5
(3) O_2^{-2} B.O = 1
(4) S_2 B.O. = 2

76)

Lattice energy ∝ (change on ion) / (distance between ions); height chargers and smaller ionic radii → higher lattice energy?
NaCl: monovalent ions → relatively low lattice energy.
$MgCl_2$: molecular lattice energy.
CaO: Ca^{2+} and O^{2-} → high energy due to +2/–2 changes.
Al_2O_3: Al^{3+} and O^{2-} → very high lattice energy (maximum among given).
Correct order: Al_2O_3 > CaO > $MgCl_2$ > NaCl.

77) K_2S > ZnS > CuS (Ionic character)

78)

Solution: Four species do not exist: Be_2, Ne_2 (Bond Order = 0); BF_6^{3-} (Boron lacks d-orbitals for hypervalency); and PH_5 (unstable hydride).
Final Answer: Option (3)

79)

d- electron in Fe^{+2} is 6
p-electron in Cl is 11

80) $E_4 = \frac{E_0}{n^2}$
$E_0 = E_4 \times n^2 = -25 \times 4^2 = -400$
PE = 2T.E. = –2 × 400 = –800

81)

$E_3 - E_2 = E(eV)$ or $\frac{-E_1}{9} + \frac{E_1}{4} = E$

 $E_1 = \frac{36E}{5} = 7.2E$

82)

 $v = 2.19 \times 10^6 \times \frac{z}{n} m/s$
z = 1 (hydrogen atom)
n = 1
$v_{1H} = 2.19 \times 10^6$ m/s, $r_{1H} = 0.529 \times 10^{-10}$ m
∵ $2\pi r = n\lambda$
$2\pi(r_1)_H = 1 \times \lambda \Rightarrow \lambda = 3.32 \times 10^{-10}$ m

83) $E = n\frac{hc}{\lambda}$
$n = \frac{E\lambda}{hc} = \frac{2 \times 2500 \times 10^{-10}}{6.6 \times 10^{-34} \times 3 \times 10^8} = \frac{50 \times 10^{18}}{6.6 \times 3} = 2.5 \times 10^{18}$

84) **Explain question:** To find out uncertainly in position if uncertainly in speed given.
Concept: Heisenberg Uncertainty Principle.
Solution:
$\Delta v = v \times \%$ error
$= 10^{-6} \times \frac{0.0001}{100}$
$= 10^{-12}$ cm sec^{-1}
$m = 10^{-10}$ g
$\Delta x \bullet m\Delta v = \frac{h}{4\pi}$
$\Delta x = \frac{h}{4\pi m} \times \frac{1}{\Delta v}$
$= \frac{0.527 \times 10^{-27}}{10^{-10}} \times \frac{1}{10^{-12}}$
$= 5.27 \times 10^{-6}$ cm
$\approx 5.2 \times 10^{-8}$ m
Final answer: Option (1)

85) NCERT, Pg.# 58

86)

NCERT-XI, Pg. # 57

87)

$\Delta x.\Delta P = \frac{h}{4\pi}$ ....(1)

we have given $\Delta P = 3\Delta x \Rightarrow \Delta x = \frac{\Delta P}{3}$

put in (1)

$\Delta P.\ \frac{\Delta P}{3} = \frac{h}{4\pi}$

$\Delta P^2 = \frac{3h}{4\pi}$

$\Delta P = \sqrt{\frac{3h}{4\pi}} = \frac{1}{2}\sqrt{\frac{3h}{\pi}}$

$m\Delta v = \frac{1}{2}\sqrt{\frac{3h}{\pi}}$

$\Delta v = \frac{1}{2m}\sqrt{\frac{3h}{\pi}}$

88) m = 0 for 5 electrons in s-orbital and $2e^-$ in p-orbital.

89) O.A.M = $\sqrt{\ell(\ell+1)}\hbar = \sqrt{0(0+1)}\hbar = 0$

90) $\frac{0.529(2)^2}{0.529(5)^2} = \frac{4}{25}$

BIOLOGY

91)

NCERT-XI, Pg. # 94

92)

NCERT-XI, Pg. # 102

93)

NCERT-XI, Pg. # 90, 100, 102

94)

NCERT-XI, Pg. # 95, 97, 98

95) NCERT-XI Pg. # 95,96

96)

NCERT-XI, Pg. # 123-124

97) NCERT-XI, Page No. 167

98)

NCERT-XI, Pg. # 121, 126

99)

NCERT-XI, Pg. # 121 – 124

100)

NCERT-XI, Pg. # 128

101) NCERT Pg. # 106, 109, 110.

102) NCERT, Pg. # 107
vinblastin is an anti cancerous drug

103) NCERT Pg.#106.

104) NCERT Pg. # 105,108,111,112 (E/H)

105) NCERT Pg. No. # 106

106) NCERT Pg. No. # 107
Cholesterol is insoluble in water

107) NCERT Page No.# 116

108) NCERT Pg. # 115

109)

NCERT-XI, Pg. # 109

110)

NCERT-XI, Pg. # 128

111)

NCERT-XI, Pg. # 127

112) NCERT-XI, Pg. # 99, 100

113) NCERT-XI, Pg. # 93

114) NCERT Pg.#106,108

115)

NCERT-XI, Pg. # 109

116) NCERT Pg. # 100

117) NCERT Pg. # 95

118)

NCERT-XI, Pg. # 96 – 98

119)

NCERT-XI, Pg. # 91

120) NCERT Pg. No. # 168

121) NCERT (XI) Pg #127

122) NCERT-XI, Pg. # 134

123)

NCERT XI, Pg. # 164, 167

124) NCERT-XI, Pg. # 127

125)

NCERT XI Pg. # 166/Module

126) NCERT-XI, Pg # 164, Last Para.

127) NCERT-XI, Pg # 164, Para.-IInd

128) NCERT Pg. # 122

129)

NCERT-XI, Pg. # 105

130) NCERT, Pg. # 106

131) NCERT Pg. # 108

132) NCERT-XI, Pg. # 145

133) NCERT Pg. # 145

134) NCERT XI Pg. No. : 118

135) NCERT Pg. No. # 115
altered structural states are representative of transition state

136)

NCERT-XI, Pg. # 198

137) NCERT-XI, Pg. # 194

138)

NCERT-XI, Pg. # 199

139) NCERT Pg # 194

140) NCERT Pg.# 194, 193

141) NCERT Pg. # 146

142) NCERT XI Pg. # 279

143)

NCERT-XI, Pg. # 196

144)

NCERT-XI, Pg. # 200

145)

NCERT XI Pg. # 285

146) NCERT-XII, Pg. # 287, 288

147) NCERT-Pg : 195-196

148)

NCERT-XI, Pg. # 200

149) NCERT Page No. 195-196

150) NCERT-Pg : 194-195

151)

NCERT Pg # 198

152) NCERT–XI, Pg # 195

153) NCERT Pg. No. # 203

154) 194, 195, 196, 197

155)

NCERT-XI, Pg. # 187

156)

NCERT-XI Pg. # 186-187

157)

NCERT XI, Pg # 189

158) NCERT-XI, Pg. # 270, 271

159)

Statement (iii) and (iv) related with inspiration

160)

NCERT-XI, Pg. # 190

161)

NCERT-XI, Pg. # 187

162) NCERT XI Pg # 268

163)

NCERT Pg. # 275

164)

NCERT-XI, Pg. # 189

165)

NCERT-XI, Pg. # 189

166)

NCERT-XI, Pg. # 188

167) NCERT (XI^{th}) Pg. # 275, Para-4

168)

NCERT-XI, Pg. # 209, 210

169)

NCERT-XI, Pg. # 205

170) NCERT (XI) Pg. # 298

171) NCERT-XII, Pg. # 294

172)

NCERT-XI, Pg. # 205

173) NCERT-XI ; Page No. # 297

174)

NCERT-XI, Pg. # 206

175) NCERT-XII, Pg. # 205, 206

176) NCERT Pg. # 206

177)

NCERT-XI, Pg. # 208

178)

NCERT, Pg # 293

179)

NCERT-XI, Pg. # 208

180) NCERT Pg # 208, 209